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The Measurement Protocol Decides the Conclusion: A Methodological Study of CNN and Vision Transformer Adversarial Robustness Comparisons

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I. INTRODUCTION

Deep neural networks are vulnerable to adversarial examples, carefully crafted perturbations that cause confident misclassification while remaining bounded by a small norm budget [1], [2]. As Vision Transformers (ViTs) [3], [4] joined convolutional neural networks (CNNs) [5] as a dominant architecture family, a natural question followed: which family is more robust, and why?

The literature answers this question inconsistently: Bai et al. [6] report that under matched training recipes transformers do not surpass CNNs, whereas Benz et al. [7] report ViTs to be more robust for standardly trained models; studies of cross-architecture transfer likewise disagree on whether attacks move more easily from CNNs to ViTs or the reverse [8]–[10]. These disagreements are usually attributed to differences in models, training recipes, or datasets, and the standard response is to compare under a matched recipe.

This paper makes a different observation, which is that even when the models, the data, the threat model, and the attack budget are all held fixed, the conclusion still depends on a choice that is rarely reported: the measurement protocol. Transfer studies must decide which samples enter the denominator (all test samples, those the target classifies correctly, those both models classify correctly, or those on which the source attack succeeded), and robustness comparisons must decide whether to report an absolute score or to separate it into clean accuracy and conditional susceptibility. These choices are usually treated as reporting details. We show that they are part of the result.

Our setting is deliberately narrow so that the measurement question is isolated. We adversarially train ResNet-18 and ViT-Tiny on CIFAR-10 [11] under a matched objective and threat model, with three independent training runs per architecture and a fixed validation split held out before any training stage, so that no model is selected on data it has seen. All evaluations run on the full test set with per-sample outcome logging, which makes paired statistics possible: the same samples can be re-weighted under different protocols without re-running an attack. We then vary the protocol, not the models, and measure how far the conclusion moves.

We also treat three claims that recur in architecture-comparison papers as measurable hypotheses, not as interpretation: that CNN input gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that ViT-specific transfer attacks improve cross-architecture transfer. Each is tested directly with a purpose-built measurement.

Our first contribution is to quantify protocol sensitivity. On identical models and data, the measured cross-architecture transfer asymmetry ranges from +4.4 to +19.4 points across four conditioning protocols, a 4.4-fold spread, whereas retraining both architectures from scratch moves the same quantity by under 1.5 points. We then separate that spread instead of leaving it as one number: 9.21 points of it separate the unconditioned rate from the target-correct rate and an identity we derive removes them in closed form, while 5.79 points come from a protocol that estimates a different quantity. What survives the correction is 3.91 points of discretion between two protocols that are both defensible, still more

than twice the retraining effect. The protocol decides whether an equivalence test is accepted or rejected, while the direction of the asymmetry, unlike its magnitude, is stable as long as the two models differ in clean accuracy; once they are matched, two of the four protocols reverse it. Repeating the full design on CIFAR-100 leaves the mean per-seed protocol spread at 13.83 ± 1.30 points, close to the CIFAR-10 value, and preserves the sign in all twelve measurements. Holding a training trajectory fixed and varying only the offline checkpoint-selection rule moves reported PGD-10 accuracy by 1.58 to 2.85 points, which places selection alongside seeds, attacks, and protocols as a reportable source of variance.

We then show that unconditioned transfer rates are confounded, and measure by how much. In a 3×3 transfer matrix that adds a WideResNet-28-10 reference, the deviation of raw from conditioned rates is not an empirical correlation but an identity. Because samples the target already misclassifies are essentially never rescued by the perturbation (0.989 to 1.000 over 36 directions), the raw rate equals the clean error plus the conditioned rate scaled by one minus that error, to within 0.41 points. Incoming transfer also tracks the target's own vulnerability, not the source architecture ($r = 0.986$ across three targets), which reframes the asymmetry as a property of the weaker target, not of stronger CNN-crafted attacks. A conditional decomposition completes the picture: absolute robustness factors exactly into clean accuracy and conditional susceptibility, and reporting both matters, because the attribution between them changed between our earlier single-run checkpoints and three fresh runs while the absolute ordering did not.

Three of the results we report are negative: scale-invariant sparsity differences do not correspond to spatial localization differences, on all four spatial measures; CLS attention entropy does not change measurably under attack at $n = 1000$; and Token Gradient Regularization [12], a ViT-specific transfer attack, gives no transfer gain in this regime, losing to plain MI-FGSM under a matched budget. Two behavioral differences do survive. Gradient sparsity differences reproduce across independent checkpoint sets, are created by adversarial training and not by architecture, and are accompanied by an alignment difference that exists only in absolute cosine. Adversarial damage accumulates at qualitatively different depths in the two architectures, and the measured drift profile depends on the token aggregation used to compute it.

We close with a short set of reporting requirements that follow from these measurements.

A preliminary version of this study was submitted to ATEEC 2026 [13]. That version reports the conditional decomposition, the four-protocol transfer comparison, the gradient structure measurements and the block-wise drift profiles on CIFAR-10 alone, and describes them as preliminary. The present paper adds two further datasets, a second threat model, the algebraic identity of Section IV, the checkpoint-selection ablation, the transformer-specific transfer attack, and the pre-registered analysis protocol, and it corrects one claim of the earlier version: the direction of the asymmetry is not stable once the architectures are matched in clean accuracy.

Section II reviews adversarial evaluation practice and architecture-comparison studies. Section III describes the training protocol, the conditioning protocols, and the measurements. Section IV reports results. Section V discusses implications and limitations, and Section VI concludes.

II. RELATED WORK

A. ADVERSARIAL ATTACKS

Adversarial attacks aim to find minimal perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] generates perturbations in a single step by following the sign of the loss gradient, with the perturbation magnitude bounded under the L_∞ norm. Projected Gradient Descent (PGD) [14] extends FGSM through iterative optimization, taking multiple gradient-sign steps and projecting back onto the ϵ-ball after each step (formal definitions in Section III-C).

The Carlini & Wagner (C&W) attack [15] formulates adversarial example generation as an optimization problem, often achieving higher success rates but at increased computational cost. More recently, AutoAttack [16] combines multiple attack strategies (APGD-CE, APGD-T, FAB-T, Square Attack) to provide a reliable robustness evaluation, becoming the de facto standard for benchmarking.

B. DEFENSE MECHANISMS

Adversarial training (AT) [14] remains the most effective defense, training models on adversarial examples generated during each batch:

$$\min_{\theta} \mathbb{E}_{(\mathrm{x},\mathrm{y})\sim\mathcal{D}} \left[\max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(\mathrm{f}_\theta(\mathrm{x}+\delta), \mathrm{y}) \right] \tag{1}$$

TRADES [17] provides a theoretically-grounded trade-off between natural accuracy and robustness:

$$\mathcal{L}_{\mathrm{TRADES}} = \mathcal{L}_{\mathrm{CE}}(\mathrm{f}(\mathrm{x}), \mathrm{y}) + \beta \cdot \mathrm{KL}(\mathrm{f}(\mathrm{x}) \| \mathrm{f}(\mathrm{x}_{\mathrm{adv}})) \tag{2}$$

where β controls the trade-off parameter.

MART [18] addresses the issue of misclassified examples by emphasizing them during training. Other approaches include input transformation [19], certified defenses via randomized smoothing [20], and ensemble methods [21]. Recent advances in certified robustness include Adaptive Randomized Smoothing [22], which improves certification radii through multi-step defenses.

C. CNN ROBUSTNESS

Extensive research has characterized the adversarial robustness of CNN architectures. Madry et al. [14] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [23] tracks state-of-the-art CNN robustness, with WideResNet variants achieving the highest robust accuracy on CIFAR-10.

Studies have shown that model capacity correlates with achievable robustness [14], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the role of skip connections in adversarial transferability [24] and the distinct batch-normalization statistics that adversarial training induces at scale [25]; relatedly, auxiliary batch-normalization branches allow adversarial examples to improve clean accuracy [26]. Adversarial training is also known to reshape input-gradient structure itself, yielding more perceptually aligned [27] and sparser [28] gradients.

D. VISION TRANSFORMER ROBUSTNESS

Vision Transformers [3], building on the self-attention mechanism introduced by Vaswani et al. [4], process images as sequences of patches; their internal representations differ measurably from CNNs' layer by layer [29]. Initial studies suggested that ViTs exhibit some inherent robustness properties due to their global receptive field [30]. Bai et al. [6] systematically investigated whether transformers are more robust than CNNs, finding that under fair training recipes transformers are not more adversarially robust than CNNs, while exhibiting superior robustness under distribution shift attributed to self-attention; Benz et al. [7] reported an early direct comparison including MLP-Mixer and linked the differences to frequency preferences. Our work extends this line of inquiry by analyzing adversarially trained models of comparable scale and providing behavioral characterizations through gradient characteristics and feature degradation analysis. Subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [8], with patch-targeted attacks exploiting the attention mechanism directly [31] and attention shown to shift measurably under patch perturbations [32].

Shao et al. [33] demonstrated that adversarial training can improve ViT robustness and that ViT features rely less on high-frequency content, though challenges remain due to training instability. ViT robustness to common corruptions and occlusions has also been characterized [34]. More recently, Jain et al. [35] proposed Adaptive Attention Scaling Adversarial Training (AAS-AT) to address ViT-specific training challenges, achieving state-of-the-art robustness on CIFAR-10 and ImageNet-100; with diffusion-augmented adversarial training, ViTs have even been reported to surpass WideResNets on small-scale datasets [36]. Zhu et al. [37] strengthened transferable attacks on ViTs through gradient normalization scaling and high-frequency adaptation. Despite these advances, behavioral comparisons between CNN and ViT robustness under a single evaluation protocol remain limited.

E. TRANSFER ATTACKS

Adversarial examples often transfer between models, enabling black-box attacks [38]. The transferability depends on factors including model architecture, training data, and attack methods [39]. Foundational transfer studies already evaluate transferability on correctly classified samples [40], and standard attack papers condition their success rates for the same reason [41]; recent evaluation guidelines make such protocol choices explicit [42]. Two recent benchmarks move in the same direction: TA-Bench [43] standardizes the comparison of more than thirty transfer attacks over a common pool of substitute and victim models, and Yu et al. [44] report that transferability is systematically overestimated when it is measured across a single architecture family, and propose three evaluation protocols that differ in how a multi-target outcome is scored: treating an ensemble as a single target, counting how many targets are fooled, and requiring that all of them are. TA-Bench holds the scoring rule fixed and varies the attack and the model pool; Yu et al. vary the scoring rule along the target-aggregation axis. We vary it along the conditioning axis instead, with the models, the data and the attack budget fixed, which isolates the contribution of the conditioning set and makes it possible to state the arithmetic relation between the resulting estimates. In ensemble design, Ravikumar et al. [9] restrict transfer measurement to samples correctly classified by both models precisely to account for accuracy differences between them, the same confound our conditioned metric controls.

Cross-architecture transfer between CNNs and ViTs has received increasing attention. Mahmood et al. [8] measured cross-architecture transfer with a both-correct protocol and found notably low CNN↔ViT transferability; Naseer et al. [10] showed that adversarial examples crafted on ViTs transfer poorly under conventional attacks and proposed self-ensemble and token refinement to improve ViT-sourced transferability, and Wei et al. [45] improved ViT-sourced transfer by bypassing attention gradients and patch-wise dropout. Building on these, Token Gradient Regularization (TGR) [12] crafts transferable adversarial samples by exploiting ViT's attention components, while Momentum Integrated Gradients (MIG) [46] achieves improved transferability across both ViTs and CNNs. Chen et al. [47] systematically analyze the factors affecting cross-architecture transferability, finding that architectural inductive biases, with CNNs preferring high-frequency information while ViTs favor low-frequency patterns, shape transfer success.

Waseda et al. [48] separate transfer outcomes further, by whether the target reproduces the source's exact error or fails in a different way, and report that both occur even between closely related models. These transfer patterns bear directly on ensemble design and on black-box threat assessment.

F. RESEARCH GAP

While individual studies have examined CNN and ViT robustness separately, and the pitfalls of unfair CNN and transformer comparisons have been highlighted before [6], [49], existing comparative work typically reports robustness scores in isolation, often with inconsistent perturbation budgets, attack configurations, or model scales, rather than pairing the comparison with behavioral analysis. A recent study in this journal compares ViT and CNN adversarial robustness directly [50]; our work differs by evaluating on the full test set with AutoAttack and paired statistics, by controlling transfer measurement for clean-accuracy differences, and by accompanying the comparison with gradient- and feature-level behavioral analyses.

The evaluation-methodology literature has repeatedly shown that how robustness is measured decides what is concluded: Carlini et al. [51] catalogue the failure modes of robustness evaluation, and Tramèr et al. [52] show that defenses judged robust under fixed attacks fall to adaptive ones. That work concerns attack strength, that is whether the measurement is strong enough. The question we ask is orthogonal: with attack strength held fixed, how much does the conditioning set alone move the conclusion? Yu et al. vary the scoring rule along a different axis, and we know of no controlled estimate along this one. Our work engages that question directly and makes it the object of study. Conditioning transfer evaluation on correctly classified samples is established practice [8], [9], [40], [41], and individual papers pick one protocol and proceed. What is missing is a controlled estimate of how much that pick matters. We supply it by holding the models, the data, the threat model, and the attack budget fixed and varying only the protocol, and we find that the measured asymmetry moves by 15.01 ± 0.84 points across the four protocols, of which 3.91 points separate the two protocols we endorse, while retraining both architectures moves that same quantity by under 1.5 points. We further show that the confound in unconditioned rates is not a tendency but an identity ($\mathrm{r_{raw}} - \mathrm{r_{cond}} = \mathrm{e}\,(1 - \mathrm{r_{cond}})$, verified to within 0.41 points over 36 directions), which turns a known qualitative caveat into a correction that can be applied directly to published results.

Three further claims recur in this literature that our design lets us test rather than assume: that CNN gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that transformer-specific transfer attacks [10], [12] improve cross-architecture transfer. We measure each directly and none holds in our setting; an untested assumption of this kind is exactly what a protocol-sensitive comparison cannot afford.

III. METHODOLOGY

A. THREAT MODEL

We consider the standard L_∞ threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(\mathrm{x}) = \{\mathrm{x}' : \|\mathrm{x}' - \mathrm{x}\|_\infty \leq \epsilon\} \tag{3}$$

Following established conventions for CIFAR-10 [11], [23], we use $\epsilon = 8/255 \approx 0.031$ as the primary perturbation budget, with additional full-test PGD-10 evaluations at $\epsilon \in \{2/255, 4/255, 16/255\}$ for the epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

B. MODEL ARCHITECTURES

a: CNN Models

ResNet-18 [5]: A widely-used baseline CNN with residual connections, containing 11.2M parameters. We use the standard architecture designed for CIFAR-10 (initial 3×3 convolution instead of 7×7).

WideResNet-28-10 [53]: A wider variant of ResNet with 36.5M parameters, representing the state-of-the-art architecture for adversarial robustness on CIFAR-10. We use the pretrained robust checkpoint of Gowal et al. [54] distributed via RobustBench [23]; note that this model was trained with additional unlabeled data and a stronger recipe than our standard adversarial training, so it serves as a reference point rather than a matched comparison.

b: Vision Transformer Models

ViT-Tiny: We use the ViT-Tiny architecture from the `timm` library [55], adapted for CIFAR-10 by bilinearly upsampling inputs to 224×224. The architecture uses 16×16 patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. The model is trained from scratch on CIFAR-10 (no ImageNet pretraining) with basic augmentation (random crop and horizontal flip), then adversarially fine-tuned. Since ViT robustness on small datasets is known to be sensitive to the training recipe [56], [57], our architecture-level conclusions should be interpreted as holding under this baseline training recipe and not for optimally trained ViTs.

C. ATTACK METHODS

a: FGSM

Single-step attack with step size equal to the perturbation budget:

$$\mathrm{x}_{\mathrm{adv}} = \mathrm{x} + \epsilon \cdot \mathrm{sign}(\nabla_{\mathrm{x}} \mathcal{L}_{\mathrm{CE}}(\mathrm{f}(\mathrm{x}), \mathrm{y})) \tag{4}$$

b: PGD

Iterative attack with step size $\alpha = 2/255$ and random initialization:

$$\mathrm{x}^{\mathrm{t}+1} = \Pi_{\mathcal{B}_\epsilon(\mathrm{x})} \left(\mathrm{x}^{\mathrm{t}} + \alpha \cdot \mathrm{sign}(\nabla_{\mathrm{x}^{\mathrm{t}}} \mathcal{L}_{\mathrm{CE}}(\mathrm{f}(\mathrm{x}^{\mathrm{t}}), \mathrm{y})) \right) \tag{5}$$

with $\mathrm{x}^0 = \mathrm{x} + \mathcal{U}(-\epsilon, \epsilon)$. We use PGD-10 (10 iterations) for both adversarial training and evaluation throughout this work (Table 1).

c: AutoAttack

We use AutoAttack [16], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-T (Auto-PGD with targeted DLR loss), FAB-T (targeted Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack). This ensemble is a standardized strong benchmark for robustness evaluation.

D. DEFENSE METHODS

a: Adversarial Training (AT)

We employ a mixed adversarial training objective that combines clean and adversarial losses:

$$\mathcal{L} = (1 - \lambda)\, \mathcal{L}_{\mathrm{CE}}(\mathrm{f}_\theta(\mathrm{x}), \mathrm{y}) + \lambda\, \mathcal{L}_{\mathrm{CE}}(\mathrm{f}_\theta(\mathrm{x}_{\mathrm{adv}}), \mathrm{y}) \tag{6}$$

where $\lambda = 0.5$ balances standard classification performance with adversarial robustness. The inner maximization generates adversarial examples using PGD-10 ($\alpha = 2/255$, $\epsilon = 8/255$), computed with the model in training mode so that attack generation and the parameter update see consistent batch-normalization statistics, following standard adversarial-training implementations [58]. This mixed objective, widely adopted in practice, helps maintain clean accuracy while building robustness, as training solely on adversarial examples can degrade natural performance. While alternative formulations such as TRADES [17] offer theoretically grounded trade-off mechanisms, this work focuses on standard adversarial training to isolate architectural effects from defense-specific interactions.

E. TRAINING PROTOCOL

a: Clean Training

For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule [59], weight decay 5×10^{-4}, and 200 epochs. For ViTs, we use AdamW optimizer [60] with initial learning rate 10^{-3}, cosine annealing, weight decay 0.05, and 200 epochs.

b: Adversarial Training

Starting from our clean-trained checkpoints, we fine-tune with reduced learning rates: CNNs use SGD with LR 10^{-3} and cosine annealing for up to 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Gradients are clipped to a global L_2 norm of 10, and batches with non-finite loss are skipped. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping; all models consume raw pixel inputs in $[0, 1]$ without mean/std normalization, and attacks and epsilon budgets operate directly in this pixel space. The training batch size is 128 for the CNN and 64 for the ViT (evaluation batch sizes vary by analysis). Early stopping uses patience 20 with a 0.1 percentage-point improvement threshold.

c: Validation Split and Repetitions

Model selection and early stopping use PGD-10 adversarial accuracy on a fixed 2,000-sample validation split. The split is drawn once (seed 777, indices stored with the code) and is shared by every model in the study, and it is excluded from the training set of both the clean and the adversarial stage, so no checkpoint is ever selected on data it was trained on. The test set is touched only by the final post-hoc evaluations. We repeat the full pipeline, clean pre-training followed by adversarial fine-tuning, three times per architecture with independent seeds (1001 to 1003 for ResNet-18, 2001 to 2003 for ViT-Tiny), and pair the runs by index so that every cross-architecture comparison is between models trained under the same repetition. All tables report the mean and standard deviation over these three runs.

For one comparison in Section IV-A we also refer to an earlier pair of checkpoints trained before this protocol was adopted, whose validation split had been part of clean pre-training. We report them explicitly as a contrast, not as results, because the difference between the two checkpoint sets is itself evidence about how far a single-run conclusion can move.

F. EVALUATION METRICS

a: Robustness Metrics

We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack). For white-box evaluations we additionally report the conditioned fooling rate, the fraction of clean-correct samples that the attack flips, and its complement, conditioned survival. Because both PGD and AutoAttack count clean-misclassified samples as non-robust in our per-sample logs, absolute robust accuracy factorizes exactly as robust = clean $\times$ conditioned survival, which lets us decompose robustness differences into a clean-accuracy component

and a conditional-susceptibility component. Finally, we report accuracies on the both-correct subset (samples that both compared models classify correctly on clean inputs), which yields an exactly paired comparison on a shared sample set.

b: Transfer Attack Metrics

Our transfer metrics are a family of conditioned fooling rates that differ only in the conditioning set. The base member is the target-correct rate: among the samples that the target model classifies correctly on clean inputs, the fraction that becomes misclassified under adversarial examples crafted on the source model. Given adversarial examples $\{x^S_{adv}\}$ generated against source model f_S, the conditioned transfer rate to target model f_T is:

$$T_c(f_S \to f_T) = \frac{\sum_{i=1}^{n} \mathbf{1}[f_T(x_i) = y_i]\, \mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]}{\sum_{i=1}^{n} \mathbf{1}[f_T(x_i) = y_i]} \tag{7}$$

Conditioning transfer evaluation on correctly classified samples follows established practice [8], [9], [40]–[42] and matters most in cross-architecture comparisons: the raw misclassification rate $\frac{1}{n}\sum_i \mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]$ includes the target's clean error, so a target with lower clean accuracy appears more "vulnerable" to transfer even when no additional samples are flipped by the attack. We report the raw rate as a secondary metric for transparency and to quantify the size of this confound. Because the conditioning set itself is a protocol choice, we additionally report two alternatives: the both-correct protocol, which conditions on samples that both the source and the target classify correctly [8], [9], and the successful-source protocol, which conditions on target-correct samples whose attack succeeded on the source in the white-box sense. Both are established. The successful-source rate is the transferability metric of Ravikumar et al. [9], whose Equation (1) divides the number of examples that transfer to the target by the number generated against the source, over a pool restricted to samples both models classify correctly; it is also the protocol implied whenever a transfer rate is reported over "successful" adversarial examples. For directional claims we take the both-correct rate as the primary estimate, because it scores both directions on identical samples and therefore supports paired inference; the other three are reported beside it throughout. For directional claims we take the both-correct rate as the primary estimate, because it scores both directions on identical samples and therefore supports paired inference; the other three are reported beside it throughout. As Section IV-B shows, these protocols can yield qualitatively different asymmetry conclusions, so we treat the protocol as part of the reported result, not an implementation detail. Transfer rates are computed on the full test set (10,000 samples) with per-sample outcomes logged, and we report 95% confidence intervals obtained by percentile bootstrap over samples (10,000 resamples).

c: Gradient Analysis Metrics

We analyze gradient properties through three complementary metric families, computing gradients of per-sample losses so that magnitudes are independent of the evaluation batch size. Gradient norm ($\|\nabla_x \mathcal{L}\|_2$ and $\|\nabla_x \mathcal{L}\|_\infty$) measures overall sensitivity magnitude. Gradient sparsity is quantified with scale-invariant measures: Hoyer sparsity, the Gini coefficient of $|g|$, and the fraction of components below 1% of the per-sample maximum, since fixed absolute thresholds conflate gradient scale with structure; Hoyer and Gini satisfy the scale-invariance axioms formalized by Hurley and Rickard [61]. Gradient alignment measures the average pairwise absolute cosine similarity across samples (computed over all pairs of the 500 analysis samples):

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i} \mathcal{L} \cdot \nabla_{x_j} \mathcal{L}}{\|\nabla_{x_i} \mathcal{L}\| \|\nabla_{x_j} \mathcal{L}\|} \right| \tag{8}$$

We use the absolute cosine because shared sensitivity axes matter for attack geometry regardless of sign; since sign coherence is what universal-perturbation arguments additionally require, we also report the signed mean cosine, which turns out to be near zero for every model (Section IV-F). High absolute alignment indicates similar perturbation axes across inputs.

d: Spatial Locality Metrics

Sparsity measures how many gradient components are large but says nothing about where they lie in the image, so we measure spatial structure separately on the channel-summed energy map $e = \sum_c (\nabla_x \mathcal{L})^2_c$, normalized to sum to one. We report the smallest pixel fraction whose energy sums to 50% and to 90% (smaller means more localized), the Shannon entropy of e viewed as a distribution over pixels (lower means more localized), and Moran's I with 4-neighbourhood weights, which is positive when energy clusters in adjacent pixels. All four are invariant to the scale of the gradient, and all are computed per sample so that the two architectures can be compared with paired tests on identical inputs.

e: Token Aggregation for Layer-wise Analysis

A transformer block emits one vector per token, so summarizing a block requires a choice. We report three: the CLS token alone, the mean over the 196 patch tokens, and the concatenation of all tokens flattened per sample. For the CNN each residual block output is flattened over its spatial dimensions. We report all three for the ViT because, as Section IV-G shows, they give different drift magnitudes and place the minimum at different depths.

f: Attention Metrics

For the ViT we extract the post-softmax CLS-to-patch attention row of each block, averaged over heads. We report its Shannon entropy, which measures how broadly attention is spread, and its displacement under attack, measured as the total variation distance between the clean and adversarial attention distributions (bounded by 1). Both are computed per sample and averaged, on n = 1000 samples per run.

g: ViT-Specific Transfer Attack

To test whether the transfer results depend on using architecture-agnostic attacks, we additionally generate source examples with Token Gradient Regularization (TGR) [12], which zeroes the extreme-valued entries of the token-gradient tensors in the attention, QKV, and MLP paths during backpropagation and scales the remaining gradient by $\gamma = 0.5$. Our implementation adapts the method to our CIFAR-10 ViT wrapper and to the budget used throughout this paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) instead of the original ImageNet defaults. Two further details of our version should be stated, because the result we report from it is negative. We apply a single $\gamma = 0.5$ to all three paths rather than tuning it per path, and we zero one maximum and one minimum entry per (sample, channel) across tokens rather than whole token rows and columns. We did not replicate the published regularization scheme exactly, so the negative result below bounds this adaptation and not the method as published; a replication with the original scheme would be needed to extend it. Plain MI-FGSM [41] examples are generated in the same run on the same samples, giving a paired control at an identical budget.

h: Feature Degradation Metrics

To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer ℓ. Feature cosine similarity measures directional preservation of the representation:

$$\mathrm{Sim}_\ell = \frac{\mathrm{h}_\ell(\mathrm{x}) \cdot \mathrm{h}_\ell(\mathrm{x}_{\mathrm{adv}})}{\|\mathrm{h}_\ell(\mathrm{x})\|\|\mathrm{h}_\ell(\mathrm{x}_{\mathrm{adv}})\|} \tag{9}$$

where h_ℓ denotes the representation at layer ℓ. Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|\mathrm{h}_\ell(\mathrm{x}_{\mathrm{adv}})\| - \|\mathrm{h}_\ell(\mathrm{x})\|}{\|\mathrm{h}_\ell(\mathrm{x})\|} \times 100\% \tag{10}$$

Feature L_2 distance ($\|\mathrm{h}_\ell(\mathrm{x}) - \mathrm{h}_\ell(\mathrm{x}_{\mathrm{adv}})\|_2$) quantifies absolute deviation.

G. STATISTICAL VALIDATION

We separate four sources of variance and report each where it applies. Training-run variance is the one that bears on architecture comparisons: every table reports the mean and standard deviation over three independent runs per architecture (Section III-E). Attack-initialization variance is measured by re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed subset; the evaluated samples and their order are fixed, so the seeds vary only the random start of the attack. Protocol variance is the spread across conditioning protocols, which we treat as a result and not as noise (Section IV-B). Checkpoint-selection variance is the spread produced by the offline rule that picks the final checkpoint from a fixed trajectory, namely validation split, early-stopping patience, and curve smoothing, measured over eighteen combinations while the trajectory itself is held constant. Section IV-I compares their magnitudes.

Paired tests are used wherever the same samples are scored under two conditions: McNemar's exact test for paired binary outcomes, Wilcoxon signed-rank with Holm correction for the three sparsity measures, paired bootstrap confidence intervals (10,000 resamples) for differences, and a sign-flip permutation test (20,000 permutations) for the both-correct transfer difference. Two units of inference are available throughout and they answer different questions. A per-sample test asks whether two fixed checkpoints differ on the images evaluated; a run-level test, over the three repetitions, asks whether the architectures differ. Only the second supports an architectural claim, and with three runs it is much the weaker. We name the unit next to every test, use the run level wherever the claim is architectural, and report both units wherever they disagree instead of the one that favors the claim; Section IV-F is where they disagree. Equivalence claims use two one-sided tests (TOST) at a primary margin of ± 2 points, fixed before the comparisons from the run-to-run variation in PGD accuracy. That is the right scale for an absolute accuracy and the wrong one for the transfer asymmetry, whose run-to-run standard deviation is 0.23 points, so there the margin is some eight times too generous and biases TOST toward accepting equivalence. We keep the pre-registered margin instead of retuning it after seeing the data, and report every equivalence claim at ± 1 and ± 3 points as well; the one equivalence we accept, on SVHN, holds at ± 1 too. All evaluation and analysis scripts fix random seeds (seed 42, including CUDA determinism flags), so every reported number is reproducible from the released pipeline.

IV. EXPERIMENTS

A. ROBUSTNESS COMPARISON

Table 1 presents the main robustness comparison across all evaluated models. Every cell for our models is the mean over three independent training runs per architecture, with the standard deviation over those runs; each run is evaluated on the full 10,000-sample test set.

The CNN holds an absolute robustness advantage under every attack: ResNet-18 AT reaches $37.93 \pm 0.14\%$

TABLE 1. Main robustness results on CIFAR-10 ($\epsilon = 8/255$), mean±std over three independent training runs per architecture. AA denotes AutoAttack with the standard suite.

Model	Defense	Params	Clean	FGSM	PGD-10	AA
CNN Models						
ResNet-18	None	11.2M	95.25±0.15	40.29±1.77	0.03±0.03	–
ResNet-18	AT	11.2M	85.78±0.36	53.46±0.08	44.11±0.50	37.93±0
WRN-28-10	AT†	36.5M	89.48	70.91	66.92	62.76
Vision Transformer Models						
ViT-Tiny	None	5.7M	80.09±0.60	6.10±1.47	0.05±0.05	–
ViT-Tiny	AT	5.7M	73.53±0.55	36.31±0.42	32.69±0.22	29.14±0

†Robust checkpoint of Gowal et al. [54] via RobustBench [23]; a single external checkpoint, so no standard deviation applies. Clean/FGSM/PGD-10: our local full-test evaluations; AA: the RobustBench-reported figure (we did not rerun AutoAttack on it).

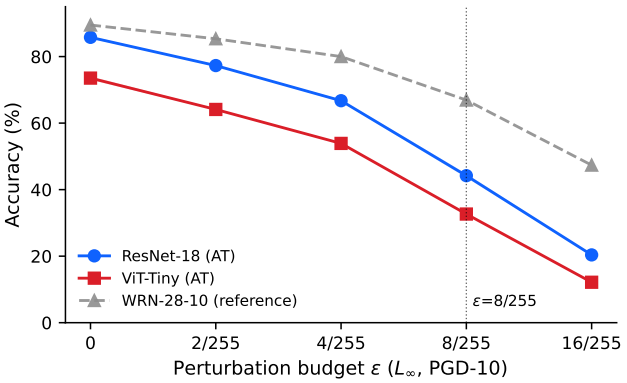


FIGURE 1. PGD-10 robust accuracy vs. perturbation budget (ϵ), measured on the full test set. Both AT architectures degrade with similar shapes as epsilon increases, with the CNN maintaining a consistent margin.

TABLE 2. Conditional decomposition of white-box robustness on the full test set, mean±std over three runs. The lower block repeats the same quantities on our earlier single-run checkpoints.

	Cond. fooling (%)		Both-correct robust (%)	
Model	PGD-10	AA	PGD-10	AA
ResNet-18 AT	48.58±0.80	55.78±0.23	59.86±0.35	51.89±0.49
ViT-Tiny AT	55.53±0.50	60.37±0.33	46.11±0.59	41.15±0.39
Earlier single run, validation split seen during clean pre-training (n=7,260)				
ResNet-18 AT	52.15	58.16	54.92	48.15
ViT-Tiny AT	52.33	56.46	49.61	45.33

robust accuracy under AutoAttack against 29.14±0.40% for ViT-Tiny AT. Because both models of a pair are evaluated on the same 10,000 samples with per-sample outcomes logged, McNemar's paired test applies within each run and favors the CNN decisively in all three pairs ($p < 10^{-84}$ under AutoAttack, $p < 10^{-120}$ under PGD-10). Seed-level variation for the adversarially trained pair is at most 0.55 points on any reported accuracy, so the ordering is not run-to-run noise; the undefended baselines vary more under FGSM (up to 1.8 points), but no claim rests on them. The CNN also retains higher clean accuracy (85.78 vs 73.53%, a 12.3-point gap), reflecting the difficulty of training ViTs on small datasets from scratch; Table 2 separates that contribution from conditional susceptibility. AutoAttack falls below PGD-10 for both architectures (6.2 points for ResNet-18, 3.6 for ViT-Tiny), confirming the value of the stronger ensemble. The WideResNet-28-10 reference [54], at 62.76%, shows how far capacity, extra training data and a stronger recipe jointly push robustness beyond the architectural choice; we keep it separate from the matched pair throughout.

Table 2 decomposes the absolute scores into their clean-accuracy and conditional-susceptibility components (Section III-F). Because misclassified clean inputs count as non-robust, the factorization is exact: for AutoAttack it reads $85.78 \times 0.442 = 37.93$ for the CNN and $73.53 \times 0.396 = 29.14$ for the ViT. Both factors favor the CNN. It starts from a 12.3-point higher clean accuracy and loses a smaller fraction of what it solves, with conditional fooling of 48.58 versus 55.53% under PGD-10 and 55.78 versus 60.37% under AutoAttack. On the $n = 7{,}061 \pm 68$ samples both models classify correctly on clean inputs, an exactly paired comparison, the CNN leads by 13.8 and 10.7 points (McNemar $p < 10^{-67}$ in every run; the bound is computed on this subset and is weaker than the full-test-set bound quoted above, which is what the larger denominator buys).

The decomposition is worth reporting even when the two factors agree, because they need not, and because their relative weight is less stable than the ordering they produce. Our first checkpoints, selected with a validation split that clean pre-training had already seen, gave the same absolute ordering but the opposite conditional one: fooling rates that were indistinguishable under PGD-10 (52.15 versus 52.33%) and slightly ViT-favorable under AutoAttack (58.16 versus 56.46%), supporting the reading that the CNN advantage was entirely a clean-accuracy effect (lower block of Table 2). None of the three corrected runs reproduces it, and they agree with one another far more closely than with the earlier run. We cannot isolate a cause, since the corrected protocol changed validation handling and the training run together; a data-quantity explanation does not hold, because the corrected clean models are better despite seeing 2,000 fewer images (80.09 ± 0.60 versus 77.50% for the ViT). The defensible conclusion is narrower and more useful than a mechanism: an attribution of this kind should not be drawn from a single training run, and a paper that reports only the absolute score hides which factor is doing the work.

Figure 1 shows PGD-10 robust accuracy across perturbation budgets $\epsilon \in \{2/255, 4/255, 8/255, 16/255\}$ on the full test set.

B. TRANSFER ATTACK ANALYSIS

This section contains the paper's central measurement. We craft PGD-10 adversarial examples on a source

TABLE 3. Transfer asymmetry under the unconditioned rate and three established conditioning protocols (PGD-10, ϵ=8/255, full test set; mean±std over three seed pairs). The rightmost column repeats the difference measured on our earlier single-run checkpoints.

Protocol	CNN→ViT	ViT→CNN	Diff.	Diff.‡
Unconditioned (raw)	41.02±0.55	27.45±0.27	+13.57±0.33	+8.27
Target-correct	19.87±0.18	15.51±0.59	+4.36±0.44	+0.63
Both-correct	18.25±0.17	9.98±0.31	+8.27±0.23	+5.33
Successful-source	36.39±0.76	17.02±0.52	+19.37±1.27	+11.17

‡Single run, validation split seen during clean pre-training.

model and evaluate them on a target, on the full test set, for each of the three seed pairs. Because outcomes are logged per sample, the same attack run can be scored under several conditioning protocols without recomputation: the protocols differ only in which samples enter the denominator (Section III-F).

The denominators differ sharply between protocols. For CNN→ViT and ViT→CNN respectively they are 10,000/10,000 without conditioning, 7,353/8,579 under target-correct, 7,061 shared samples under both-correct, and 2,831/3,814 under successful-source. The spread in the Diff. column is the standard deviation of the paired within-seed difference, which the two per-direction spreads cannot recover because the directions covary across seeds; it is the training-run reference of Section IV-I. The protocol, not the model, dominates the measured quantity. On identical models and identical data, the asymmetry ranges from $+4.36 \pm 0.44$ to $+19.37 \pm 1.27$ points; the mean per-seed spread between the largest and the smallest protocol estimate is 15.01 ± 0.84 points, a factor of 4.4, whereas the seed-level standard deviation within any single protocol stays below 1.5 points. That spread is not a single quantity, and separating it is part of the result. Take the unconditioned rate as the pivot: the successful-source rate sits 5.79 points above it and conditions on the source's own white-box failure, so it estimates a different quantity, while the target-correct rate sits 9.21 points below it and that entire gap is what Equation (11) predicts in closed form, so a reader given the target's clean error can remove it exactly. The two account for the full 15.01 points. What survives the correction is the difference between the two protocols we endorse for directional claims, target-correct and both-correct, which differ only in whether the source must also be correct: 3.91 points, still more than twice what retraining both architectures moves. That residual is 5.97 points on CIFAR-100 and 0.65 on SVHN. The pivot structure is dataset-dependent: on CIFAR-100 the unconditioned rate is the largest of the four, so the whole 13.58-point spread there is the identity term and the successful-source rate falls inside it. The protocol also decides the statistical verdict. Under the both-correct protocol, which scores both directions on the same samples and therefore supports paired inference, the difference is 8.27 ± 0.24 points, differing in the third decimal from the ± 0.23 of the table only because the two estimators differ. Each run has its own paired bootstrap CI ($[7.61, 9.44]$, $[7.10, 9.01]$ and $[7.28, 9.20]$); we summarize them by the average of their endpoints, $[7.33, 9.22]$, which is a description of three intervals and not itself an interval with a coverage guarantee, and we give the across-run t interval, $[7.68, 8.86]$, for the inference that is actually at the run level. The sign-flip permutation p is below 5×10^{-5} in every seed, and TOST unable to establish equivalence at any of the three margins (± 1, ± 2 and ± 3 points, $p = 1.0$ in each), the interval lying entirely outside ± 3; under the target-correct protocol on our earlier checkpoints, the same pipeline accepted equivalence at ± 2 points. A reader given only one of these numbers would draw a different conclusion about whether the architectures differ at all.

A protocol name is not by itself a specification. The successful-source protocol conditions on samples whose attack succeeded on the source, and whether that phrase requires the source to classify the clean input correctly is a sub-choice that transfer studies rarely state. We report the stricter reading, in which white-box success means that the source is correct on the clean input and wrong on its adversarial counterpart. Under the looser reading, which also admits samples the source already misclassified, the measured asymmetry falls from +19.37 to +14.60 points and the protocol spread from 15.01 to 10.45 points. That shift of 4.77 points is larger than the run-to-run standard deviation of any protocol in Table 3 by more than a factor of three, and the other three protocols are unchanged by construction. Because a sub-choice of this size is invisible in a protocol's name, all four conditioning masks are defined in a single place in the released code.

What is stable is the direction: CNN-crafted examples transfer better to the ViT than the reverse under all four protocols, in all three seeds, and also under the momentum-based MI-FGSM [41] (19.07 ± 0.33 versus $15.42 \pm 0.52\%$ target-correct; this bidirectional check uses the attack's own default step of $\alpha = \epsilon/10$, not the $\alpha = 2/255$ used elsewhere, which is why its ViT→CNN figure differs slightly from the matched-budget value in Table 11), so the effect is not an artifact of the PGD attack itself. The magnitude, its significance, and any equivalence claim are protocol-dependent; the sign is not.

The mechanism behind the spread is compositional. Samples that the target solves on clean input but the source does not are three to four times more fragile than jointly solved ones ($41.2 \pm 0.8\%$ and $59.1 \pm 1.7\%$ fooling versus 10.0 and 18.3%), and these marginal subsets differ five-fold in size ($1{,}517 \pm 102$ versus 291 ± 16) because the two models differ in clean accuracy. Conditioning on target-correct admits both marginal subsets, whose asymmetric sizes pull the two rates toward

each other; conditioning on both-correct excludes them, and the directional difference reappears. This account covers three of the four protocols. The successful-source protocol is driven differently, and separating the two drivers changes what the spread depends on: across 18 direction pairs drawn from six seed-and-dataset clusters, the spread of the four-protocol asymmetry falls as the clean-error gap between the two models widens (slope -0.528, cluster-bootstrap CI $[-0.664, -0.418]$), whereas the spread computed without the successful-source protocol rises with it (slope $+0.431$, CI $[+0.333, +0.633]$). The successful-source rate is the extreme of the four in 16 of the 18 pairs, and the widest pair is target-correct against successful-source (23.77 points average gap, widest in 13 of the 18 directions). Conditioning on source success selects by the source's own robustness, which is not a function of the pair's clean-accuracy gap, so it contributes a second and partly opposing term. The pool for this fit excludes the SVHN pair, and we record how that exclusion arose because it is not what the pre-registration says. The registration placed SVHN in the fitted arm and named it as the pair that would occupy the empty band below twelve points on the x axis; its reassignment to an independent consistency check was written after the SVHN result was in hand. We therefore treat the SVHN-inclusive fit as the pre-registered one. Adding its two directions moves the four-protocol slope to -0.133, with a cluster-bootstrap CI of $[-0.570, +0.469]$ that includes zero, while the three-protocol slope keeps its sign and its interval ($+0.470$, CI $[+0.361, +0.665]$). We therefore withdraw the four-protocol slope; the mechanism claim rests on the three-protocol one, which survives the pool question and is what Section IV-I carries forward.

We calibrate the protocol spread against the clean-error gap in two ways that fail differently, and they agree on three of the four protocols. The controlled arm holds one member of a pair fixed and sweeps the other along its own training trajectory, so architecture, seed, dataset, and attack budget stay constant while the gap moves (116 points from eight seed-and-dataset clusters). The observational arm compares final models across architectures, seeds, and datasets, where everything moves at once (18 direction pairs from six clusters). Inference in both is by cluster bootstrap at the trajectory level; the degrees of freedom are set by the number of clusters, not by the number of points, and we do not pool the arms. Two cautions apply to these intervals. With six to eight clusters a percentile cluster bootstrap undercovers, so the intervals below should be read as lower bounds on width, not as exact. And the WideResNet is a single fixed external checkpoint that appears in all six observational clusters, in twelve of the eighteen direction pairs, so those clusters are not fully independent; the pair-only sensitivity that excludes it is reported alongside.

Excluding the successful-source protocol, the two arms agree: $+0.673$ (CI $[+0.602, +0.727]$) in the controlled arm against $+0.470$ (CI $[+0.361, +0.665]$) in the observational arm on the pre-registered pool, same sign and overlapping intervals. The four-protocol slope, which we withdrew above, supports no comparison either way: on the pre-registered pool the observational interval $[-0.570, +0.469]$ contains the controlled arm's $+0.273$ (CI $[+0.219, +0.371]$) entirely, so the arms neither agree nor disagree there in any informative sense. We record the post-hoc contrast only to show what the pool choice does. Dropping SVHN moves the observational four-protocol slope to -0.528 (CI $[-0.664, -0.418]$), which does not overlap the controlled arm, and restricting that pool to the matched architecture pair moves it to -0.100 (CI $[-0.464, +0.092]$); a quantity this sensitive to which pairs enter the fit is not one a mechanism claim can rest on. The agreement that does carry the claim is the three-protocol one, and it holds in every pool we fit. Two limits belong with this: the observational arm's negative slope is fitted across an unmeasured gap of 7.2 points on the x axis, so it compares two separated clusters instead of tracing a trend; and the controlled arm's four-protocol slope keeps its sign but not its magnitude when early checkpoints are dropped, moving over $+0.201$ to $+0.492$ across thresholds, a range its own interval of $[+0.219, +0.371]$ does not cover; the three-protocol slope is the most stable quantity of all (0.661 to 0.673).

1) How much of the raw rate is the target's clean error?

To quantify the confound directly instead of arguing it, we extend the analysis to a 3×3 matrix by adding the WideResNet-28-10 reference as both source and target (Table 4). The relationship between the two rates turns out to be stronger than a correlation: it is an identity resting on a single empirical fact.

Write e for the target's clean error, r_{raw} for the unconditioned rate and r_{cond} for the target-correct conditioned rate. If a sample the target already misclassifies on clean input is still misclassified after the perturbation, then $r_{raw} = e + r_{cond}(1 - e)$, and therefore

$$r_{raw} - r_{cond} = e\,(1 - r_{cond}). \tag{11}$$

The antecedent is the part that is measured, and it holds tightly: across 36 off-diagonal directions (two datasets, three seed pairs, three targets, two sources) the probability that a clean-misclassified sample is also misclassified under attack ranges from 0.989 to 1.000. Equation (11) consequently reproduces every measured unconditioned rate to within 0.41 points, with a mean absolute residual of 0.095 points; the residual accounts for at most 1.39% of the deviation being explained.

This changes the status of the claim, and strengthens it. The deviation of unconditioned from conditioned

transfer rates does not merely correlate with the target's clean error in our data; it is that error, scaled by $(1 - r_{cond})$, up to a residual below half a point. Unconditioned transfer rates are contaminated by target weakness by construction, in any study whose targets satisfy the antecedent, not as a regularity we happened to observe on three targets. The antecedent is itself a measurable quantity and should be measured rather than assumed. Writing p for the probability that a clean-misclassified sample stays misclassified under attack, the general form is $r_{raw} = e\,p + r_{cond}\,(1-e)$, which reduces to Equation (11) when $p = 1$. Where $p < 1$, Equation (11) overstates the deviation by exactly $e\,(1 - p)$, so it is an upper bound and not an equality once the antecedent fails. We measure p between 0.989 and 1.000 on adversarially trained targets, where the correction is therefore exact to within a few tenths of a point. On standard-trained targets, which is the regime most transfer studies attack, we observed p as low as 0.88 in our own pre-correction runs; we quote that as an order-of-magnitude indication and not as a measured value of the present pipeline, and there the identity is a bound. The algebra itself is not new. Written over a single model instead of a source-target pair, $r_{raw} = e + r_{cond}(1-e)$ is the natural-plus-boundary decomposition of TRADES [17], where the premise corresponding to our p holds by definition because the boundary term is defined by an existential quantifier over the perturbation neighborhood. What is new here is that in the transfer setting the premise is not definitional: a fixed source-crafted perturbation either preserves the target's clean error or does not, so p becomes an empirical quantity, and we measure it rather than assume it. We previously reported this as a Pearson correlation ($r = 0.997$), which understated it: a near-unit correlation between a quantity and its own dominant component is arithmetic, and the small-sample caveat it required is unnecessary once the identity is written down.

What remains dataset-specific is the proportionality factor, not the mechanism. Because $(1 - r_{cond})$ depends on how transferable the attack is, the fitted slope is 0.762 points per point of target error on CIFAR-10 and 0.656 on CIFAR-100. It is an average over the targets measured, not a constant, and it should be re-estimated, not assumed, when the identity is applied to other model pools. Unconditioned transfer rates therefore do not measure transferability across architectures; they measure transferability plus the target's clean weakness, in a proportion that Equation (11) makes explicit.

Figure 2 shows the conditioned matrix. The third architecture also reframes the asymmetry itself. Averaged over the two foreign sources, incoming conditioned transfer is 18.50% for the ResNet target, 19.04% for the ViT target, and 10.57% for the WideResNet target, which tracks each target's own white-box conditioned fooling rate (48.52, 55.57 and 25.16%) with $r = 0.986$ over the three targets. On this evidence the CNN→ViT advantage is better read as a property of the weaker target than as evidence that CNN-crafted perturbations are intrinsically more transferable: the most robust target absorbs attacks from both families almost equally well. With only three targets this correlation is suggestive rather than conclusive, and we report it as such.

TABLE 4. **3 × 3 transfer matrix (PGD-10, ϵ=8/255, full test set; mean±std over three seed pairs; the WideResNet column and row use the single external checkpoint). Raw = unconditioned fooling rate, Cond. = target-correct conditioned rate. Diagonal entries are white-box attacks.**

Source	Target	Raw (%)	Cond. (%)
ResNet-18 AT	ResNet-18 AT	55.84±0.53	48.52±0.84
ResNet-18 AT	ViT-Tiny AT	41.07±0.58	19.93±0.20
ResNet-18 AT	WRN-28-10	21.69±0.09	12.48±0.10
ViT-Tiny AT	ResNet-18 AT	27.45±0.34	15.52±0.64
ViT-Tiny AT	ViT-Tiny AT	67.34±0.18	55.57±0.44
ViT-Tiny AT	WRN-28-10	18.25±0.12	8.65±0.14
WRN-28-10	ResNet-18 AT	32.63±0.14	21.48±0.31
WRN-28-10	ViT-Tiny AT	39.73±0.25	18.15±0.28
WRN-28-10	WRN-28-10	33.03	25.16

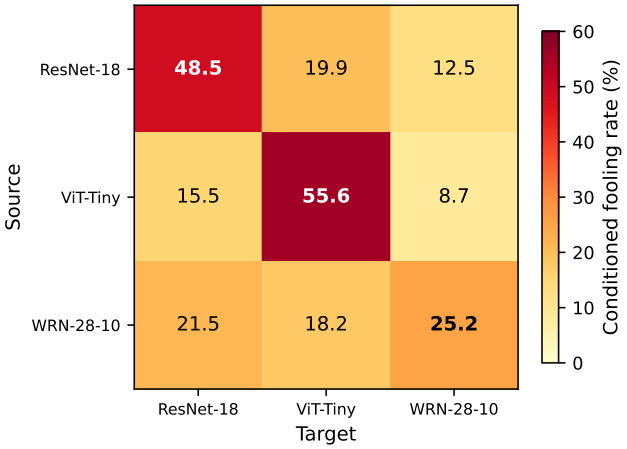


FIGURE 2. **Conditioned (target-correct) transfer fooling rates, mean over three seed pairs. Off-diagonal asymmetry under this protocol is +4.4 points; Table 3 shows how far the same quantity moves under other protocols.**

C. DOES THE PROTOCOL EFFECT GENERALIZE? A SECOND DATASET

The measurement above is made on one dataset, and a protocol effect that exists only on CIFAR-10 would not support a general reporting recommendation. We therefore repeated the whole design on CIFAR-100: same architectures, same threat model, same attack budget, same three-seed structure, same four protocols. We registered three directional predictions and three endpoint bands in an append-only document before any measurement from this dataset existed, so that the outcome could not be reinterpreted after the fact. The registration does not precede the start of training, and we say so rather than let "pre-registered" be read as

TABLE 5. CIFAR-100 replication (mean±std over three independent training runs per architecture, full test set, ϵ=8/255). Both models are adversarially trained with the protocol of Section III-E. Absolute robustness is lower than on CIFAR-10 for both architectures, and the CNN-over-ViT ordering is unchanged.

Model	Clean (%)	PGD-10 (%)	AutoAttack (%)
ResNet-18 AT	63.86±1.05	19.30±0.32	15.04±0.54
ViT-Tiny AT	43.17±1.10	11.15±0.60	8.87±0.83

preceding the run: the registration commit precedes the first reported number, an epoch-5 pilot gate, by sixty-six minutes.

Across the same four protocols the measured CNN→ViT asymmetry runs from $+4.96 \pm 1.01$ points (target-correct) to $+18.53 \pm 0.71$ points (unconditioned), with $+10.92 \pm 0.34$ under both-correct and $+17.50 \pm 0.92$ under successful-source. The mean per-seed spread is 13.83 ± 1.30 points, against 15.01 ± 0.84 on CIFAR-10, so the harder dataset leaves the protocol effect in the same range, slightly below the CIFAR-10 value instead of above it. The direction is positive in all twelve measurements (four protocols × three seed pairs), as it was on CIFAR-10. Under the both-correct protocol the paired difference is 10.92 points. As on CIFAR-10, the quoted interval $[9.47, 12.35]$ is the average of the three per-seed bootstrap endpoints rather than an interval with a coverage guarantee; the across-run t interval, which is the one that matches a claim about architectures, is $[10.07, 11.77]$. The sign-flip permutation p is below the resolution of 20,000 permutations in every seed, and TOST does not establish equivalence at any margin tested.

One of the three registered predictions is confirmed, one fails under the reading we treat as binding, and the third is not testable in this design. The prediction that the sign would be preserved is confirmed (12/12). The prediction that the protocol spread would be larger on the harder dataset was registered as a comparison against the CIFAR-10 spread, quoted at the time as 10.45 points. Against that literal threshold it holds ($13.83 > 10.45$); against the CIFAR-10 spread as it now stands under the successful-source definition of Section IV-B it does not ($13.83 < 15.01$). We report the failure rather than the reading that would rescue it, because the registration named the CIFAR-10 spread and not the number. The two datasets place the protocol effect in the same range instead of ordering it by difficulty, so the mechanism we propose predicts the size of the effect but not its ordering between these two datasets. The third predicted that the deviation of the unconditioned from the conditioned rate would grow with the target's clean error. The measured slope is +0.656 points per point of target error, positive as predicted; but on CIFAR-100 two of the three targets have clean errors that differ by 0.21 points (36.15 and 36.36), so the correlation is computed over two effective clusters, not three, and its value ($r = 0.931$ over six directions, 0.974 aggregated to targets) rests on a two-point contrast. The measurement is consistent with the prediction and does not test it; we report it as such and not as a third confirmation.

One registered endpoint was missed: we predicted clean accuracy in the 33 to 40% band for the adversarially trained ViT and measured 43.17 ± 1.10%. The band was derived by assuming that the clean-accuracy cost of adversarial training scales the same way for both architectures. It does not: the ViT retained 0.885 of its clean pre-training accuracy under adversarial training, against 0.802 for the ResNet. The two adversarial endpoints, which are the ones the protocol claim depends on, fell inside their bands (PGD-10 predicted 10 to 14%, measured 11.15; AutoAttack predicted 8 to 11%, measured 8.87).

Because conditioning changes which samples enter the denominator, it also changes the class mix, so an asymmetry could in principle be a composition artifact; it is not. We decompose each direction's rate into a composition component and a rate component. Under target-correct the composition effect is −1.249, −1.330 and −1.065 points across the three seed pairs, that is 17.4 to 29.7% of the asymmetry and negative: rebalancing the class mix would make the asymmetry larger, not smaller. Under successful-source it is small and positive in every seed (+0.527, +0.435 and +0.382 points, that is 2.1 to 3.1% of the asymmetry). Under the unconditioned and both-correct protocols it is exactly zero, which is the arithmetic check the decomposition must pass.

Two qualifications belong with this result. The two adversarial bands held with margins of 0.22 points (AutoAttack) and 0.48 points (PGD-10) at the nearest seed-level edge. Those margins are smaller than the checkpoint-selection noise on this same dataset: sweeping early-stopping patience and curve smoothing over the CIFAR-100 trajectories, with the training run held fixed, moves the reported PGD-10 accuracy by 0.18 to 1.83 points (mean 0.86). The same nine-cell sweep on CIFAR-10 gives 0.54 to 2.83 points per validation split, and the eighteen-cell figure quoted in Section IV-I is larger still because it also varies the split, a dimension the CIFAR-100 protocol does not have. The bands held, but by margins at or below the selection noise of the dataset they were registered for; a reader should weigh the pre-registration accordingly. Second, the same held-out split serves selection at both stages instead of two independent splits. That choice is inherited deliberately from the CIFAR-10 protocol so the two datasets remain comparable, and Section IV-I reports what it does and does not cost.

TABLE 6. SVHN transfer asymmetry under the four protocols (PGD-10, ϵ=8/255, full test set n=26,032; mean±std over two seed pairs; two architectures only, so no third-architecture control). Unlike CIFAR-10 and CIFAR-100, the sign is not the same in all four rows.

Protocol	CNN→ViT	ViT→CNN	Diff.
Unconditioned (raw)	31.89	31.49	+0.39±0.08
Target-correct	26.32	27.33	−1.00±0.22
Both-correct	25.36	25.70	−0.35±0.01
Successful-source	61.38	58.69	+2.70±0.40

TABLE 7. Same checkpoints, L_2 budget (ϵ=0.5; mean±std over three seed pairs). PGD-L2 uses the full test set; AutoAttack-L2 uses a pre-registered 5,000-sample subset, so its clean column differs slightly from the full-set value and the two are not drawn from the same samples.

Model	Clean	PGD-L2	Clean (5k)	AA-L2
ResNet-18 AT	85.78	63.78±0.38	85.33	60.61±0.42
ViT-Tiny AT	73.53	51.72±0.23	73.61	49.85±0.20

D. A THIRD DATASET WHERE THE ARCHITECTURES ARE MATCHED

CIFAR-10 and CIFAR-100 both place the two architectures far apart in clean accuracy (about 12 and 21 points). SVHN [62] does not (Table 5 reports the CIFAR-100 absolutes for comparison): adversarially trained under the same recipe, the ResNet reaches $94.27 \pm 0.70\%$ clean and the ViT $92.42 \pm 0.56\%$, a gap of 1.85 points. Robust accuracies are 55.52 ± 0.64 and 52.30 ± 0.10 under PGD-10. This is the regime our thesis makes the sharpest prediction about, and the prediction is not the one a reader of the previous sections would expect.

When the architectures are matched, the protocol decides the sign. Two of the four protocols report a CNN advantage and two report a ViT advantage (Table 6). The split is not arbitrary. The two protocols we endorse for directional claims, target-correct and both-correct, agree with each other and both favor the ViT; the sign change separates them from the unconditioned rate and the successful-source rate. The unconditioned reversal is moreover the arithmetic consequence of Equation (11): the ViT target's clean error exceeds the CNN's by about two points here, which is enough to carry a -1.00 conditioned difference to $+0.39$ raw. Reporting a raw rate can therefore flip the sign of a published comparison, and the identity says in advance when it will. Under the both-correct protocol, which we treat as primary because it scores both directions on identical samples, the difference is -0.35 points, with a sign-flip permutation p of 0.105 and TOST equivalence accepted at all three margins, including the tightest one we test (± 1 point, $p = 0.0007$). One qualification belongs with that acceptance. SVHN was run with two seed pairs, so the quoted interval $[-0.76, +0.07]$ averages the endpoints of two per-seed bootstraps and no across-run interval is available: with two runs the run-to-run standard deviation is estimated from a single contrast and carries no information. The equivalence therefore rests on the within-run comparison, and we report it because the two runs agree closely (-0.34 and -0.36), not because two runs bound the run-to-run variation. A study reporting only the successful-source rate would announce a 2.70-point CNN advantage on SVHN; one reporting only the target-correct rate would announce a 1.00-point ViT advantage. Both would be arithmetically correct and neither would be reproducible by the other.

This behavior is consistent with the mechanism, not an exception to it. The protocol spread on SVHN is 3.70 ± 0.62 points, against 15.01 ± 0.84 on CIFAR-10 and 13.83 ± 1.30 on CIFAR-100, roughly four times smaller than the two CIFAR values while the clean-accuracy gaps are 1.85, $\approx$ 12 and $\approx$ 21 points. The relation is not monotone across the three: the largest gap does not produce the largest spread, and CIFAR-100 sits slightly below CIFAR-10. What separates the datasets here is the near-matched pair, not the ordering among the mismatched ones. Once the spread is small enough it no longer clears the asymmetry itself, so the sign becomes a property of the protocol and not of the architectures. The direction we reported as stable in Section IV-B is stable in the regime where the two models differ in clean accuracy, which is the regime both CIFAR datasets happen to occupy.

SVHN was run in a reduced configuration fixed in advance: two seed pairs instead of three, 50 adversarial-training epochs, no AutoAttack, and no third architecture. The last of these means the 3×3 confound control of Section IV-B1 cannot be built here, and we do not build a within-SVHN version of it.

E. DOES THE PROTOCOL EFFECT SURVIVE A CHANGE OF THREAT MODEL?

Everything so far is measured inside one threat model. To test whether the protocol effect is an L_∞ artifact, we re-evaluated the same CIFAR-10 checkpoints under an L_2 budget of $\epsilon = 0.5$, registering three directional predictions before looking at any L_2 output. The models are L_∞-trained, so this measures the measurement axis, not L_2 robustness, and the absolute values below are not comparable with L_2-trained references.

Two of the three registered predictions hold and the third turns out not to be testable. The protocol spread is 10.92 points under L_2 against 15.01 under L_∞, so it is smaller under the changed norm but far above both the two-point floor the pre-registration set and the run-to-run standard deviation, and the sign is positive in all twelve measurements. The third prediction, that the deviation of unconditioned from conditioned rates would rise with the target's clean error, cannot fail once Equation (11) holds: the deviation is $e\,(1 - r_{\text{cond}})$, so it rises with e by construction, and the fitted slope of

TABLE 8. Gradient characteristics computed from per-sample loss gradients (n=500 test samples per run; mean±std over three runs).

Metric	ResNet-18 AT	ViT-Tiny AT	ViT (native)‡
Hoyer sparsity	0.4928±0.0120	0.4561±0.0055	0.405
Gini coefficient	0.6498±0.0089	0.6099±0.0051	0.571
Frac. <1% of max	34.8%±2.1	26.8%±0.9	17.9%
Alignment (absolute)	0.0378±0.0009	0.0562±0.0004	0.079
Alignment (signed)	0.0014	0.0005	–

‡CIFAR-native ViT control (4×4 patches, 5.4M params); a single checkpoint predating the fixed-split protocol, selected on the test set, included only for the sign of the upsampling effect.

$+0.914$ estimates $1 - r_{cond}$ rather than testing anything. In any case we report the slope rather than a correlation, because the eighteen directions take only seven distinct target-error values, six of them the same fixed WideResNet. We registered it before writing the identity down; we now report it as an out-of-sample check on the identity, which is the status we also give it in Section IV-C. The four protocols (Table 7) report asymmetries of $+13.03 \pm 0.81$ (unconditioned), $+2.12 \pm 0.18$ (target-correct), $+4.07 \pm 0.36$ (both-correct) and $+12.06 \pm 0.23$ (successful-source). Absolute robustness keeps the CNN-over-ViT ordering under both attacks, with McNemar $p < 10^{-57}$ in every seed pair.

That check passes. Conditioned rates are far lower under this L_2 budget than under L_∞ (roughly 2 to 9% against 9 to 22%), so the identity predicts a steeper slope, and the measured 0.914 sits where it should, against 0.762 on the same models under L_∞. The proportionality factor is a property of the attack budget, not a constant, exactly as Section IV-B1 states.

F. GRADIENT CHARACTERISTICS

To characterize architectural differences in input sensitivity, we analyze the gradient properties of both architectures, together with a CIFAR-native ViT control (32×32 inputs, 4×4 patches, no upsampling) trained with the same adversarial objective but before the fixed-split protocol described in Section III-E was adopted, which isolates the effect of the 32→224 upsampling in our main ViT pipeline. Table 8 summarizes the key statistics using scale-invariant sparsity measures (Section III-F); gradients are computed with per-sample losses so that norms do not depend on the evaluation batch size. Alignment there is the mean absolute cosine similarity over all $500 \times 499/2$ sample pairs, and every ResNet to ViT difference on the scale-invariant sparsity metrics is significant under per-sample paired Wilcoxon tests in every run, with Holm-corrected $p < 10^{-11}$ and paired Cohen's d between 0.32 and 1.23.

Across all three scale-invariant sparsity measures, CNN gradients are sparser than ViT gradients, and the ordering reproduces in every run with seed-level standard deviations at least three times smaller than the gap. This is the one place where our two units of inference disagree, so we report both. Within a run, where the same 500 samples are measured under both models, per-sample paired tests are decisive on all three measures in all three runs (Wilcoxon signed-rank, Holm-corrected $p < 10^{-11}$ throughout, paired Cohen's d between 0.32 and 1.23); that establishes that these checkpoints differ on these images. At the run level, which is the unit an architectural claim needs and the unit Table 9 uses, three paired runs give raw p of 0.036, 0.042 and 0.067 for Gini, the sub-1% fraction and Hoyer, which Holm correction across the three measures raises to 0.108. The ordering is consistent three runs out of three on every measure and the gap is several times the run-to-run standard deviation, but with three runs it does not clear $\alpha = 0.05$ after a correction that is conservative here because the three measures are near-collinear. We therefore report the sparsity result as reproducible and not as significant. The alignment difference below does reach run-level significance ($p = 0.0009$), and it is the gradient result we carry into the discussion on that basis. The ordering is a product of adversarial training, not of architecture: in the clean-trained counterparts of the same runs the ViT is the sparser model (Hoyer 0.380 ± 0.009 versus 0.347 ± 0.002), so adversarial training sparsifies CNN gradients far more strongly, consistent with [28]. The native-resolution ViT control is less sparse than the upsampled ViT on every measure, so the ordering is not an artifact of the $32 \rightarrow 224$ upsampling. Cross-sample alignment is higher for the ViT in absolute cosine, and the clean-trained pair already shows this gap (0.058 versus 0.029), which makes it a property of the architecture and not of the defense. The signed mean cosine is near zero for both models (≤ 0.0014): what samples share is sensitivity axes, not consistent directions, so the alignment gap is not evidence for cheap universal perturbations.

These gradient results are also the one part of our analysis that is insensitive to which checkpoint set is used: every quantity in Table 8 reproduces the ordering measured on our earlier single-run checkpoints, although in six of the eight cases the value itself moves by more than the seed-level standard deviation reported here. The contrast with the conditional decomposition of Section IV-A is therefore about direction, not magnitude: there the attribution between clean accuracy and conditional sensitivity changed between checkpoint sets, whereas here no gradient-structure difference changes sign.

1) Sparsity is not spatial locality

Sparsity measures answer how many gradient components are large, not whether those components sit together in the image. The distinction matters because the sparser-CNN result invites a spatial reading, in which gradients are taken to be more "localized" or

TABLE 9. Spatial locality of input gradients (n=500 samples per run; mean±std over three runs). Paired differences are computed on the same samples. Inference is at the run level: p is a one-sample t-test on the three per-run mean differences, Holm-corrected across the four measures, so that the unit of inference matches the unit of the claim. Lower area and entropy mean more localized; higher Moran's I means more clustered.

Measure	ResNet-18	ViT-Tiny	Diff.	p
Area for 50% energy	0.0378±0.0017	0.0397±0.0010	−0.0019	0.54
Area for 90% energy	0.2345±0.0078	0.2570±0.0067	−0.0224	0.40
Spatial entropy (nat)	5.2014±0.0387	5.2524±0.0197	−0.0509	0.54
Moran's I	0.4141±0.0132	0.3941±0.0024	+0.0200	0.47

"concentrated" on object regions. We do not attribute that reading to a particular study; we tested it because it is the natural interpretation of a sparsity result and because our own earlier figures encouraged it. The measures do not support it. We therefore measure spatial structure directly on the same gradients, using four scale-invariant descriptors of the channel-summed energy map: the smallest pixel fraction carrying 50% and 90% of the gradient energy, the spatial entropy of the normalized energy map, and Moran's I with 4-neighborhood weights, which is positive when energy clusters in adjacent pixels.

The result is negative (Table 9). None of the four measures shows a significant difference at the run level. The largest of them, the area carrying 90% of the energy, points toward a smaller area for the CNN by two percentage points of image area and does not survive correction. Compared at the same unit, the spatial measures are not merely weaker than the sparsity ones: the sparsity ordering is consistent in every run and every measure while the spatial ordering is not stable across runs on two of the four measures, which is the difference that matters even though neither family clears run-level correction at three runs. Per-sample tests within a single run reach much smaller p-values here too, but they answer the narrower question of whether two fixed checkpoints differ on these 500 images. Neighborhood clustering is essentially identical for the two architectures. The two families of measures therefore disagree, and the disagreement is informative: adversarially trained CNN gradients place more of their mass on fewer pixels, but those pixels are not more spatially grouped than the ViT's. Descriptions of CNN gradients as spatially localized are not supported by our measurements, and we restrict our own claims to sparsity.

Figure 3 provides visual evidence for these gradient differences.

Figure 4 shows a direct comparison of gradient heatmaps, highlighting the structural differences in how each architecture computes input gradients.

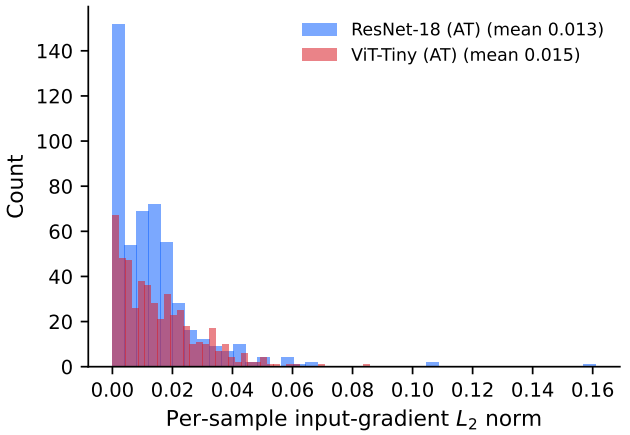


FIGURE 3. Distribution of per-sample gradient L_2 norms for both architectures. The magnitude maps for the same inputs are shown in Figure 4.

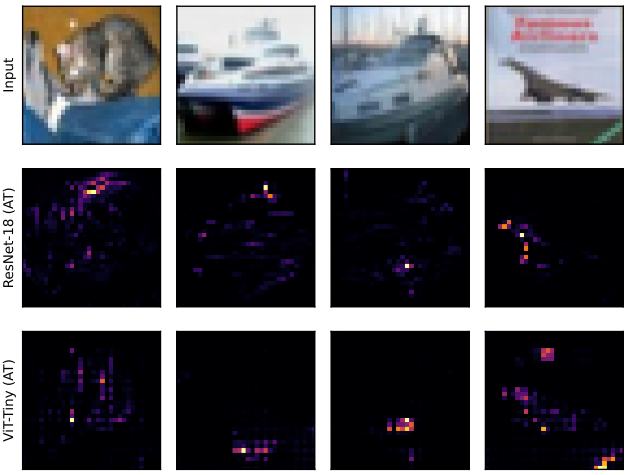


FIGURE 4. Input-gradient magnitude maps for the same test images under both models. CNN gradients place more of their mass on fewer components (Table 8); Table 9 shows that this is not accompanied by a difference in spatial arrangement, so the panels should be read as sparsity and not as localization.

G. FEATURE DEGRADATION IN VISION TRANSFORMERS

We next measure where adversarial damage accumulates along depth, extending layer-wise representation analyses of clean-trained ViTs [29] to the adversarially trained setting. Two design choices in this measurement are usually left implicit and turn out to matter, so we vary both: the sample size (n = 1000 per run instead of a single small batch) and the token aggregation used to summarize a transformer block (Section III-F).

Two results follow. The architectures differ less in whether damage accumulates than in where. In the ViT the drift is gradual and shallow: similarity declines monotonically to a minimum at block 11 and recovers slightly at the output, while feature norms stay within 0.45% of their clean values throughout, so the perturbation rotates the representation instead of

TABLE 10. Block-wise clean-to-adversarial cosine similarity under each model's own white-box PGD-10 attack (n=1000 per run; mean over three runs). All summaries are taken at the residual block output, so the ViT and CNN columns are measured at equivalent points; the ViT columns differ only in how a block's tokens are aggregated. Figures for the MLP sub-block output are given in the text.

ViT-Tiny AT				ResNet-18 AT	
Blk	CLS	Patch mean	All tokens	Block	Cosine
1	0.9993	0.9980	0.9927	layer1.0	0.9951
2	0.9950	0.9959	0.9901	layer1.1	0.9918
3	0.9902	0.9940	0.9881	layer2.0	0.9879
4	0.9873	0.9919	0.9861	layer2.1	0.9853
5	0.9845	0.9904	0.9847	layer3.0	0.9753
6	0.9814	0.9894	0.9835	layer3.1	0.9607
7	0.9782	0.9885	0.9826	layer4.0	0.8777
8	0.9743	0.9875	0.9816	layer4.1	0.9108
9	0.9703	0.9870	0.9809		
10	0.9661	0.9872	0.9805		
11	0.9604	0.9874	0.9803		
12	0.9663	0.9915	0.9837		

rescaling it. In the CNN the damage is concentrated: the first three stages lose little, then the first block of the last residual stage collapses to 0.8777 ± 0.0181 together with a $-13.14 \pm 1.89\%$ norm contraction, after which the final block partially repairs the representation (0.9108 ± 0.0060, norm $+5.84 \pm 3.53\%$). Adversarial damage in the CNN is thus a late, localized, magnitude-changing event; in the ViT it is a distributed, direction-only rotation. We note that on our earlier checkpoints the CNN profile fell monotonically to the last block, with no recovery, so the shape of the CNN curve, unlike the ViT profile, should be read as recipe-dependent.

Second, the aggregation choice changes the measured profile substantially. At their respective minima the CLS pathway drifts about three times as far as the patch-token mean (0.9604 versus 0.9870, a ratio of 3.05 in $1 - \cos$), and the two summaries place the minimum at different depths (block 11 versus block 9). Since the CLS token at the block output is what the classifier reads, this is not a nuisance but a result: the perturbation concentrates on the classification pathway while the average patch representation is comparatively preserved. The tensor the summary is taken from matters as much as the aggregation. Read at the MLP sub-block output instead of the residual block output, the same three summaries give a CLS minimum of 0.9343 at block 10, a patch-token minimum of 0.9840 at block 8 and a ratio of 4.12: the drift looks larger, the depths reorder, and the sub-block is not the tensor the classifier reads. Layer-wise drift curves from different papers are therefore not comparable unless both the aggregation and the tensor are stated.

Table 10 reports the block-wise profiles. A third measurement is negative. Adversarial perturbation does not measurably change the entropy of the CLS attention distribution: over $n = 1000$ samples per run, the mean per-layer change is at most 0.005 nats in absolute value against layer means near 4.9 nats, with a seed-level standard deviation of the same order. We state this as equivalence rather than as a failure to reject: with a margin of ± 0.05 nats, one percent of the layer mean and fixed from the scale of the measure and not from the data, TOST accepts equivalence in all twelve layers, the weakest at $p = 0.006$. What does grow with depth is the displacement of the attention map, from 0.028 total variation in the first layer to 0.084 in the last, small in absolute terms, since total variation is bounded by 1. The attack therefore redirects attention slightly without broadening or sharpening it; Figure 6 plots the per-layer curves, which lie on top of each other. We flag this explicitly because an earlier version of this analysis, computed on an 8-sample figure batch, suggested a visible attention effect that the larger measurement does not support; any claim that adversarial attack degrades attention should be checked at this sample size before being made. We raise this against our own preliminary version rather than against a specific study in the literature.

Finally, drift magnitude does not predict attack success. Splitting the samples by whether the attack flipped the prediction, the last-block cosine similarity of flipped and unflipped ViT samples is statistically indistinguishable (0.9895 versus 0.9877); for the CNN there is a small difference in the expected direction (0.9058 versus 0.9143). Feature drift is therefore a description of how the representation moves, not a sufficient statistic for whether the prediction changes.

Figure 5 visualizes CLS-token attention maps at layers 1, 6, and 12 for a representative sample. It agrees with the displacement measurement, since later layers move more, but single-sample panels of this kind are easy to over-read: the aggregate entropy change over 1,000 samples is null, so the visible differences reflect redirection and not diffusion. Attention-shift effects under patch perturbations [32] and attacks that target the attention mechanism directly [31] operate in a different regime from the L_∞-bounded perturbations studied here.

Penultimate-space geometry is measured in the original feature space, not from a low-dimensional embedding, which would invite qualitative over-reading (n=500 per run, white-box, mean±std over three runs). The attack collapses class structure most strongly in the CNN: clean-fit 5-NN label consistency drops from $87.3 \pm 2.6\%$ to $51.3 \pm 1.8\%$, the silhouette score from 0.196 ± 0.010 to -0.032 ± 0.004, and the relative class-centroid shift is 0.238 ± 0.012 against 0.083 ± 0.004 for the ViT, whose space is weakly class-separated even on clean inputs (silhouette -0.021 ± 0.005; 5-NN consistency $61.1 \pm 1.4\%$, dropping to $48.5 \pm 1.4\%$). The measurable statement is therefore that the CNN's well-separated clean feature space is heavily disrupted by white-box attack, whereas the ViT's class structure is weaker to begin with and moves less in these aggregate terms; this

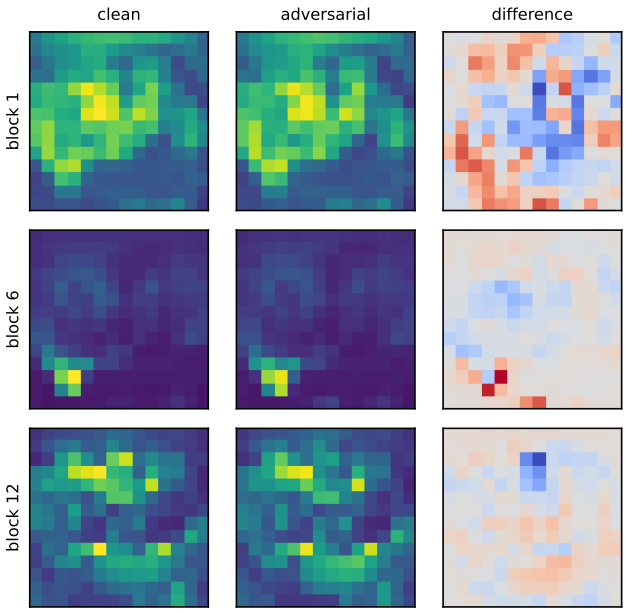


FIGURE 5. CLS-token attention maps (post-softmax, head-averaged) across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12); columns show clean attention, adversarial attention, and their difference. A single sample is shown for illustration; the depth trend in attention displacement is reported quantitatively in the text over $n = 1{,}000$ samples per run.

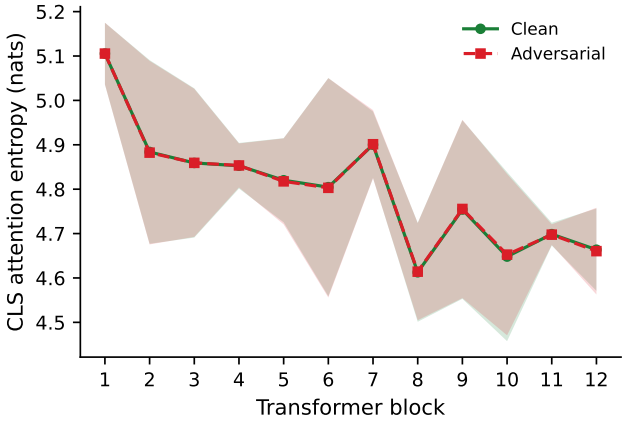


FIGURE 6. Per-layer entropy of the CLS attention distribution for clean and adversarial inputs, $n = 1{,}000$ samples per run, mean over three runs with ± 1 s.d. shading. The attention maps themselves are shown in Figure 5.

is a statement about geometry, not about robustness, since the CNN nonetheless retains the higher robust accuracy.

H. DOES A VIT-SPECIFIC TRANSFER ATTACK CHANGE THE PICTURE?

All transfer results so far use gradient attacks that treat the ViT as a generic differentiable model. A natural objection is that the ViT direction is handicapped by this choice, since attacks designed for transformer structure are reported to transfer better. We test the objection directly with Token Gradient Regularization (TGR) [12], which suppresses extreme-valued token gradients in the attention, QKV, and MLP paths during backpropagation on the source model. Our implementation is an adaptation: the original targets standardly trained ImageNet ViTs, whereas we apply the same mechanism to our CIFAR-10 ViT-Tiny wrapper at the budget used throughout this paper (Section II-A), with plain MI-FGSM generated in the same run as a paired control.

TABLE 11. ViT→CNN transfer under TGR [12] and plain MI-FGSM at a matched budget (n=10,000 per run; mean±std over three runs).

Measure (%)	TGR	MI-FGSM
White-box on source (raw)	51.56±0.80	66.43±0.29
Transfer, target-correct	12.70±0.41	15.71±0.60
Transfer, both-correct	7.93±0.42	10.08±0.30

TGR does not improve transfer in this regime; it degrades it (Table 11). The regularization behaves as designed on the source side, trading away 14.9 points of white-box strength, but the trade does not pay off at the target: conditioned transfer falls by 3.0 points under the target-correct protocol and by 2.2 under both-correct. Paired McNemar tests on the both-correct subset favor MI-FGSM decisively in every run ($p = 6.1 \times 10^{-10}$, 1.1×10^{-27}, 7.0×10^{-34}), so this is not a seed effect.

We state the conclusion narrowly, because our setting differs from the one TGR was designed for in three respects that could each matter: the targets are adversarially trained, not standardly trained, the data are 32×32 CIFAR-10 images upsampled inside the model instead of native ImageNet, and the source is a 5.7M-parameter ViT-Tiny. The claim we can support is that TGR's transfer advantage does not carry over to this regime, not that the method fails in general. For the purposes of this paper the relevant consequence is that the CNN→ViT asymmetry reported in Section IV-B is not an artifact of using architecture-agnostic attacks: giving the ViT direction a transformer-specific attack makes it weaker, not stronger.

I. SOURCES OF VARIANCE

Four sources of variance bear on the comparisons above. Attack-initialization stochasticity is negligible: re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed 1,000-sample subset leaves clean accuracy exactly unchanged, since the loader does not shuffle, and moves robust accuracy by a standard deviation of 0.05 points for the CNN and 0.00 for the ViT. Training-run variance is larger but still small: across the three independent runs per architecture, the standard deviation for the adversarially trained pair is at most 0.55 points on clean accuracy and 0.50 on PGD-10 (Table 1).

Comparing the protocol effect against those two figures directly would be an error of scale, because the protocol effect is measured on the transfer asymmetry while both figures above are measured on absolute accuracy. We therefore quantify the protocol and training-run effects on the same quantity, the off-diagonal asymmetry, and report both in absolute units. Within a fixed seed pair, changing only the conditioning protocol moves the asymmetry by 15.01±0.84 points. This figure is the mean of the per-seed ranges; it coincides with the range of the protocol means here because the extremal protocols are the same in all three seeds, and differs on CIFAR-100 (13.83 against 13.58) because there they are not. Retraining both architectures from scratch moves that same asymmetry by a standard deviation of only 0.23 to 1.27 points, the value depending on which protocol it is measured under (Table 3, Diff. column). A fifteen-point measurement effect against a sub-1.5-point training effect is the ordering the paper rests on.

We deliberately do not headline the ratio of these two spreads. With three runs the denominator carries two degrees of freedom, so a χ^2 interval on any single ratio is very wide: for the successful-source protocol it runs from 0.8 to 9.9 and includes unity. Reported as a sensitivity, the ratio spans 5.2 to 32.6 across dispersion measures and reference protocols, and equals 28.3 under the like-for-like comparison on the both-correct protocol we treat as primary. Two cautions bound it further: the protocol with the largest asymmetry also carries the largest run-to-run standard deviation, so numerator and denominator are not independent, and because all three runs share one validation split the denominator excludes split-to-split variance, which makes the ratio an upper bound. The absolute comparison survives both, which is why it and not the fold-change carries the claim. This ordering is the quantitative form of the paper's argument: with a protocol left unstated, a reader cannot tell a real architectural difference from a measurement choice.

A fourth source is the offline checkpoint-selection protocol, which is usually left implicit. Holding a training trajectory fixed and varying only how its final checkpoint is chosen (validation split, early-stopping patience, and whether the validation curve is smoothed before the argmax; eighteen combinations) moves the reported PGD-10 accuracy by 2.62 to 2.85 points per seed for the ResNet and 1.58 to 2.09 points for the ViT. The weights on disk are identical across these cells; only the rule that picks one of them differs. About half of that dispersion comes from one of the three dimensions. Smoothing the validation curve before the argmax lowers test PGD-10 accuracy monotonically in every ResNet run and in two of the three ViT runs, so it is not a neutral alternative but a systematically worse rule; restricted to the split and patience choices that studies actually vary, the dispersion falls to 1.09 times the seed-level standard deviation for the ResNet and 1.42 times for the ViT. We report both: the larger measures what an undocumented preprocessing step can add, the smaller what a documented protocol still leaves open. The dispersion is quoted in absolute units and deliberately not as a ratio to seed-level dispersion, because that ratio depends on which of the eighteen cells is the reference: across all eighteen it runs from 0.67 to 3.38 for the ResNet and 0.63 to 7.38 for the ViT, inverting the comparison for some of them.

A counterweight belongs immediately next to it: on CIFAR-100 the epoch selected by our own protocol varies by 25 epochs across the three ResNet seeds, yet the resulting test PGD-10 standard deviation is 0.32 points. The selection path was unstable while the selected outcome was not. Neither observation generalizes without the other: selection choices can move a reported number by more than a training seed does, and they can also leave it nearly untouched, and which of the two happens is a property of the trajectory's plateau and not of the architecture. The practical consequence for reporting is the same in both cases: the selection rule is part of the result and belongs in the methods section.

Computational Details: The training runs and analyses reported here were conducted on an NVIDIA RTX 5090 (32GB VRAM); the earlier single-run checkpoints used for the comparison in Table 2 were trained on an RTX 5060 Ti (16GB). All runs used PyTorch 2.6.0 with CUDA 12.8 and deterministic cuDNN settings. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on the full 10,000-sample test set in seeded 1,000-sample chunks with per-sample outcome logging. The three-seed training pipeline required 21.7 GPU-hours in total.

V. DISCUSSION

A. WHAT THE MEASUREMENTS SHOW

The central result is an ordering of dispersions: measured on the transfer asymmetry itself, retraining both architectures from scratch moves it by a standard deviation of 0.23 to 1.27 points, whereas changing only the conditioning protocol moves it by 15.01 ± 0.84 points (Section IV-I). We state the two spreads in absolute units rather than as a fold-change, because with three runs the interval on that ratio is too wide to carry a headline. For scale on a different quantity, re-running an attack with a new random initialization moves absolute robust accuracy by about 0.05 points and retraining moves the adversarially trained pair by at most 0.55. A quantity that the measurement choice moves by fifteen points while retraining moves it by one is not a stable property of the architectures, and reporting it without the protocol communicates almost nothing.

One objection deserves an answer in its own terms. The four protocols estimate four different quantities, so the distance between them is not a variance and calling

it one would be a category error. We agree, and we are not claiming that the architectures have a single true asymmetry that the protocols estimate with error. Our claim is about what a reader can recover. When a study reports one number without naming the protocol, the reader cannot tell which of the four quantities was estimated, and the spread we measure is the size of that ambiguity. It is a bound on what the omission costs, not a dispersion of a single estimator. This is not an argument against conditioning, which remains established practice [9], [40], [41] and necessary, but an argument that the choice belongs in the result and not in a footnote.

Nor is it an argument that our direction is the correct one. Under a both-correct protocol comparable to ours, Mahmood et al. [8] report the opposite direction on a different architecture pair, finding transfer into the transformer weaker than transfer out of it. Our mechanism is compatible with that result: we attribute the asymmetry to the target's own vulnerability rather than to any special strength of CNN-crafted perturbations, and the weaker target in their pair is not the weaker target in ours. Testing this properly would require their per-direction clean accuracies under a shared protocol, which is exactly the reporting we ask for and the sharpest external test the mechanism admits.

The four protocols are not interchangeable, and each has a defensible use. The unconditioned rate should not be used for architecture comparison at all: our 3×3 matrix shows its deviation from the conditioned rate is algebraically equal to the target's clean error scaled by $(1 - r_{cond})$, which means it measures target weakness in a known proportion, not transferability. The target-correct protocol removes that confound but admits samples the source itself cannot solve, and because those marginal subsets are hyper-vulnerable and asymmetrically sized, it pulls two directions toward each other by an amount that depends on the clean-accuracy gap. The both-correct protocol scores both directions on identical samples and is the only one of the four that supports paired inference, which is why we treat it as the primary estimate. The successful-source protocol answers a different question, how well an attack that already worked carries over, and its larger value reflects that conditioning on source success selects strong perturbations.

Across four protocols and three training runs, adversarial examples crafted on the CNN transfer better to the ViT than the reverse on both CIFAR datasets, and on CIFAR-10 the direction also holds under a second attack family (MI-FGSM). On SVHN, where the same recipe leaves the two architectures 1.85 points apart in clean accuracy, not 12.3 or 20.7, that is no longer true: two protocols report a CNN advantage and two a ViT advantage, and under the primary both-correct protocol the difference is -0.35 points with equivalence accepted (Section IV-D). The direction is a property of the unmatched regime, not of the architectures: while the models are unmatched the protocol changes only how large the effect looks and whether equivalence is accepted, and once they are matched it changes the sign as well. Our three-target analysis reads this direction without requiring the CNN's perturbations to be special: incoming transfer tracks each target's own white-box vulnerability ($r = 0.986$), and the most robust target absorbs both families almost equally. On this account the asymmetry is a statement about which model is the weaker target, which is consistent with our finding that a transformer-specific attack [12] does not rescue the ViT direction: with three targets the correlation is suggestive, and a larger model pool would be needed to establish it.

The conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. In our corrected runs both favor the CNN, so the decomposition and the headline agree. They did not agree on our earlier checkpoints, where conditional susceptibility appeared matched under PGD-10 and slightly ViT-favorable under AutoAttack. Since the absolute ordering was identical in both checkpoint sets, a paper reporting only absolute accuracy would have seen nothing change, while a paper reporting the decomposition from a single run would have drawn a mechanistic conclusion that does not replicate. The practical lesson is that mechanism-level claims need more training runs than ranking-level claims, and the two should not be given the same evidential weight.

Sparsity is not spatial locality: adversarially trained CNN gradients concentrate more mass on fewer components (Hoyer 0.4928 versus 0.4561, consistent in every run; per-sample $p < 10^{-11}$, run-level $p = 0.108$ after correction), but four spatial descriptors of the same gradients show no corresponding difference in how those components are arranged in the image. Descriptions of CNN gradients as spatially localized therefore need separate evidence; sparsity measures [61] do not provide it.

Adversarial perturbation does not measurably change CLS attention entropy. At $n = 1000$ the per-layer entropy change is at most 0.005 nats, while attention maps are displaced by a small and depth-increasing amount. Attention-shift results obtained with patch attacks [31], [32] concern a different threat model and should not be generalized to L_∞-bounded perturbations without measurement.

A ViT-specific transfer attack does not improve ViT-sourced transfer here. TGR [12] costs 14.9 points of white-box strength and returns none of it at the target in our regime. We report this as a regime-specific negative result, since our targets are adversarially trained and our data are CIFAR-scale, and it is exactly the kind of claim that a single untested assumption would otherwise import into a comparison.

Two behavioral differences reproduce across independent checkpoint sets and are unaffected by the protocol question. Gradient structure differs: CNN input gradients are sparser under all three scale-invariant measures, adversarial training, not architecture, creates that ordering (the clean-trained pair has it reversed), and cross-sample alignment is higher for the ViT in absolute cosine while the signed mean is near zero for both, so that samples share sensitivity axes, not directions, which is the distinction that separates our measurement from a universal-perturbation claim [63]. Depth profiles differ: adversarial damage in the CNN is a late, localized, magnitude-changing event confined to the last residual stage, whereas in the ViT it is a distributed, direction-only rotation. That the ViT's measured profile depends on token aggregation, since the CLS pathway drifts about three times as far as the patch-token mean, is itself a reporting requirement for layer-wise studies.

B. REPORTING REQUIREMENTS

The measurements above imply a short and concrete set of requirements for cross-architecture robustness studies.

Studies should state the conditioning protocol and report which samples enter the denominator. Where a directional claim is made, a protocol that scores both directions on identical samples should be preferred, so that paired inference is available. Unconditioned transfer rates should not be reported as architecture comparisons. When targets differ in clean accuracy, the raw rate is a known mixture of transferability and target weakness, $r_{raw} - r_{cond} = e\,(1 - r_{cond})$, and because this is an identity rather than a fitted relation the correction can be applied to published raw rates without re-running anything.

Studies should also report the decomposition and not only the score. Absolute robustness factors exactly into clean accuracy and conditional susceptibility, and which factor carries a headline is the interesting part, which is not recoverable from the score alone. Ranking claims should be separated from mechanism claims: our rankings were stable across checkpoint sets while our mechanism attribution was not, so mechanism claims should carry more training runs. For layer-wise analyses the aggregation should be stated, since CLS-only, patch-mean, and all-token summaries of the same blocks give different drift magnitudes and different minima. Finally, attention and locality claims should be checked at a realistic sample size, because both of the null results we report would have looked positive on a small figure batch.

For practitioners choosing between the two families under matched standard adversarial training on this benchmark, the CNN gives higher absolute robust accuracy and the advantage holds under every view we measured. The WideResNet reference at 62.76% shows that recipe, capacity, and extra data jointly [54] move robustness further than the architectural choice does, which is worth weighing before investing in architecture selection as a robustness strategy.

C. LIMITATIONS

Our study is deliberately narrow, and the narrowness bounds the conclusions in specific ways.

The protocol-sensitivity result is measured on three datasets and one model pair. On CIFAR-100 the spread stays close to the CIFAR-10 value and the sign is preserved throughout (Section IV-C). The effect is therefore not CIFAR-specific, but its size does not follow the clean-accuracy gap across datasets: CIFAR-100 has by far the larger gap and the slightly smaller spread. The mechanism we propose predicts that a gap produces a spread, not how the spread orders between datasets. That account is incomplete, and we state the limit rather than the slogan: it holds for three of the four protocols, while the successful-source protocol adds a second driver tied to the source's robustness that is strong enough to reverse the overall relationship (Section IV-B). What we still cannot state is the functional form of either dependence with confidence: the estimates rest on six seed-and-dataset clusters, and every small-gap pair in our pool involves the external WideResNet, so a small clean-accuracy gap and a different training recipe are not separable here. A study spanning several architecture pairs with varying clean-accuracy gaps would turn these measurements into a calibration curve.

We cannot isolate the cause of the differences between our two checkpoint sets. The corrected protocol changed the validation handling and produced a fresh training run at the same time. A data-quantity explanation is ruled out, since the corrected clean models are better despite less data, but distinguishing selection leakage from ordinary run-to-run variation would require an ablation that trains on identical data with and without a leaked selection split. We report the contrast as evidence that single-run mechanism claims are fragile, not as a measurement of leakage.

Every model in this paper is trained inside a single threat model: L_∞ at $\epsilon = 8/255$. We evaluated the same checkpoints under an L_2 budget and the protocol spread survived, at 10.92 against 15.01 points (Section IV-E), but that test probes the measurement axis, not L_2 robustness, and its absolute values are not comparable with L_2-trained references. Whether the effect holds for models trained under another norm is untested here.

Three runs per architecture bound seed variance but do not span the recipe space, and ViT adversarial training is known to be recipe-sensitive on small datasets [56], [57]; with diffusion-generated synthetic data ViTs have been reported to surpass WideResNets on CIFAR-10 [36], [64]. Our ViT results characterize baseline adversarial training, not the best achievable ViT robustness. The ViT-Tiny pipeline upsamples 32 ×

32 inputs to 224 × 224 inside the model; our CIFAR-native control shows the gradient ordering is not an upsampling artifact, but that control differs in capacity and patch size and therefore calibrates the confound instead of providing a second matched pair. The two models also differ in parameter count (11.2M versus 5.7M), so architectural attribution of the absolute gap remains partial. Finally, we evaluate a single-budget L_∞ threat model, and our TGR result is an adaptation of the original method, not a reproduction of its published setting.

D. FUTURE DIRECTIONS

The most direct extension is to turn the protocol-sensitivity measurement into a general one: sweep architecture pairs with controlled clean-accuracy gaps and measure how the spread across protocols scales with that gap. If the relationship is as regular as our three-target correlation suggests, it would yield a correction that lets published raw-rate asymmetries be reinterpreted rather than discarded.

Two extensions we previously listed as future work have since been run. The leakage ablation, on six full-budget trajectories, did not detect an effect of leakage on checkpoint selection, but it answers a weaker question than the one posed: the treatment was applied to a single validation split, so split identity and leakage are not separable within that design, and its informative by-product is the selection-protocol dispersion of Section IV-I. The second dataset has also been run (Section IV-C). What remains open is the capacity-matched pair (ViT-Small against ResNet-50, say), which would test whether the sparsity ordering and the depth profiles are properties of the families or of these particular small models.

VI. CONCLUSION

We asked how much of the disagreement in CNN and Vision Transformer robustness comparisons comes from the measurement and not from the models, and answered it by holding the models, the data, the threat model, and the attack budget fixed while varying only how the comparison is scored.

The measurement dominates: on identical adversarially trained ResNet-18 and ViT-Tiny pairs, the cross-architecture transfer asymmetry ranges from +4.4 to +19.4 points across four conditioning protocols, whereas retraining both architectures from scratch moves that same quantity by under 1.5 points. That 15.01-point spread decomposes into 9.21 points that an identity we derive removes in closed form and 5.79 points contributed by a protocol that estimates a different quantity; what survives the correction is 3.91 points of genuine discretion between two defensible protocols, and that is the part a reporting standard has to address. Unconditioned rates are not a valid basis for such a comparison at all: in a 3×3 matrix that adds a WideResNet-28-10 reference, their deviation from conditioned rates is algebraically equal to the target's clean error scaled by one minus the conditioned rate, an identity that holds to within 0.41 points across 36 measured directions and that applies to any published raw rate whose targets satisfy the measured antecedent. The same design run on CIFAR-100 leaves the spread at 13.83 ± 1.30 points, close to the CIFAR-10 value, and an ablation that fixes the trajectory and varies only the checkpoint-selection rule moves reported robust accuracy by 1.58 to 2.85 points, so measurement choices that are usually left implicit are of the same order as the effects being reported. The direction is stable across protocols, seeds, and attack families on both CIFAR datasets, but not across datasets: on SVHN, where the same recipe leaves the two architectures only 1.85 points apart in clean accuracy, two of the four protocols reverse the sign and the primary protocol accepts equivalence. What survives is narrower and more useful than a ranking: the direction holds while the models are unmatched, and our three-target analysis associates it with the target's own vulnerability ($r = 0.986$) and not with any special potency of CNN-crafted perturbations.

In the conditional decomposition of absolute robustness both factors favor the CNN in our corrected runs, but the attribution between them differed on an earlier checkpoint set whose absolute ordering was identical, which shows that mechanism-level claims are more fragile than ranking-level claims and need more training runs to support.

Three readings did not survive direct measurement: adversarially trained CNN gradients are sparser but not spatially more localized, CLS attention entropy does not change under L_∞ attack at $n = 1000$, and a transformer-specific transfer attack yields no transfer gain in this regime. Only the third is a published claim; the first two are natural readings that our own earlier figures encouraged, and we test them as such. Two behavioral differences did survive and reproduce across independent checkpoint sets: the gradient sparsity ordering, which adversarial training creates and not architecture, and the depth profile of adversarial damage, which is a late localized magnitude-changing event in the CNN and a distributed direction-only rotation in the ViT.

From these measurements we distilled a short set of reporting requirements, of which the first two matter most: state the conditioning protocol and prefer paired ones, and do not present unconditioned transfer rates as architecture comparisons. Our study covers three datasets and a single model pair, so the size of the protocol effect for other architecture pairs remains to be calibrated; the mechanism behind it, an unequal clean-accuracy gap between targets, is not specific to our setting. The pipeline, the per-sample logs and the analysis scripts will be released on acceptance so that the

protocol comparison can be reproduced and extended.

DATA AND CODE AVAILABILITY

The experiments in this study use the CIFAR-10 and CIFAR-100 datasets [11] and the SVHN dataset [62], all of which are publicly available. Pre-trained WideResNet-28-10 weights were obtained from the RobustBench benchmark [23], [54]. Upon acceptance we will publish the analysis pipeline, the scripts that generate every table and figure in this paper, the per-sample evaluation records, the fixed validation-split indices and the seed lists, which together reproduce every number reported here; the four conditioning protocols are all computed from the per-sample records, so the protocol comparison can be repeated without retraining. An anonymized copy of that archive is available to reviewers during review at `[ANONYMOUS-ARCHIVE-URL]`, so the protocol comparison can be checked before acceptance rather than after it. Trained checkpoints total about 40 GB and are available from the authors on request.

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